

Gowtham Ayna Raveendran

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Summary

Data Scientist specializing in **Anomaly Detection** and **Natural Language Processing**. Expert in leveraging data-driven insights to drive business outcomes and build scalable solutions to handle large-scale data processing and analysis. Skilled in deploying machine learning models using Spark with expertise in building post-deployment monitoring and observability pipelines and doing **root cause analysis** while also adept at pioneering new solutions for complex, ill-defined challenges where requirements evolve rapidly. Proficient in **SQL**, **Python** and **R** and cloud technologies (AWS) with expertise in building **Tableau** and **Looker** dashboards.

Professional Experience

Yahoo

Remote (Portland, OR)

Data Scientist II - Traffic Protection

July '21 – Present

Core Data Scientist responsible for identifying active threat vectors, monitoring system health in Advertising network traffic

- Designed, built and deployed a **device throttling pipeline for Bid traffic**, dynamically populating a denylist to temporarily restrict high-volume devices. Model resulted in a cost savings of **\$250k/year** without impacting ad spend from advertisers
- Engineered and deployed a **model interpretability and observability pipeline** for Real-time scoring model to surface prediction drivers and to detect model drift, establishing proactive model health monitoring and reducing root cause analysis time by **75%**
- Developed and Integrated a dedicated model track within the **Weight-of-Evidence based Real time traffic scoring framework** to handle ConnectedTV bid traffic, improving detection granularity and the framework's ability to handle new traffic types
- Spearheading a POC effort for the org level AI-ML Working Group, evaluating and prototyping a scalable **Retrieval-Augmented Generation (RAG) agent** using **AWS Bedrock Agentcore** to automate and streamline repetitive operational queries (e.g., Rule information, Model explanations, historical trends) for the Traffic Protection team.
- Leading a cross-functional Working Group to perform deep data profiling and trend analysis on a 3rd-party IP intelligence dataset, defining new criteria to refine **VPN and Residential Proxy traffic handling**, enhancing traffic protection by reducing fraudulent traffic in advertising and search platforms.

Yahoo

Champaign, IL

Data Scientist – Brand Understanding Group, Datalabs

Jan '18 – June '21

- Fine tuned trained language model for **Named Entity Recognition (NER)** and **targeted sentiment analysis** models. These models were hierarchically operated - NER model identifies the brands in article headlines while SA model measures sentiment towards brands; Deployed the model using PySpark processing over **100M+ records/hour**.
- Developed **Brand Churn Detection model** using Recency-Frequency-Monetary value based segmentation to create targeted advertisement segments generating **\$1M** revenue under influence. This work is **patented**.
- Designed a solution for measuring **Brand Affinity** for users for building targeted user segments based on their interest profile.

TrueScripts Management Services

Washington, IN

Data Analyst

Jun '17 – Jan '18

- Developed the company's first automated reporting system using Tableau, reducing turn around time by a factor of **10x**.
- Engineered and Automated drug repricing system using R for estimating potential savings for new clients
- Architected and Built the company's first centralized **data warehouse**, standardizing data access across departments.

Education

Indiana University Bloomington

GPA: 3.84/4.0

M.S. in Data Science

May '17

Anna University, India

GPA: 8.36/10

B.E. in Electronics & Communication

Jun '12

Skills

Machine Learning: Keras, Scikit-learn, Decision Trees, PCA, Ensemble learning - Random Forests, XGBoost, Autoencoders, One class SVMs, Isolation Forests, DBSCAN, K-Means, mlflow, glum

Tools & Technologies: Airflow, Oozie, MLflow, Docker, Git, Looker, Otel, Grafana Tableau, Terraform, Spark, Hadoop, Hive, Vertica, Presto, Athena, Amazon Bedrock Agentcore, SageMaker, EMR, Athena, BigQuery(GCP)

Languages: Python, SQL, R, Bash, Java, Pig, Scala